Cryptography

Diffie-Hellman Key Exchange and the Discrete Log Problem

Cristhian Iza

School of Mathematical and Computational Sciences
Yachay Tech University

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Discrete Log Problem

Public-Key Cryptography



UC 2.

Public-Key Cryptography and Key Establishment

- Introduction to Public-Key Cryptography
- Number Theory and Modular Inverses
- RSA Cryptosystem
- RSA Implementation and Security
- Diffie–Hellman Key Exchange and the Discrete Log Problem
- Key Establishment and Security Attacks

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Bruce Schneier



Twofish,
Blowfish

Vincent Rijmen
and Joan Daemen



AES

Modern private
key encryption

Rivest, Shamir
& Aldeman



Public-k
encrypt

Ron
Rivest



Hashing

Whitfield
Diffie



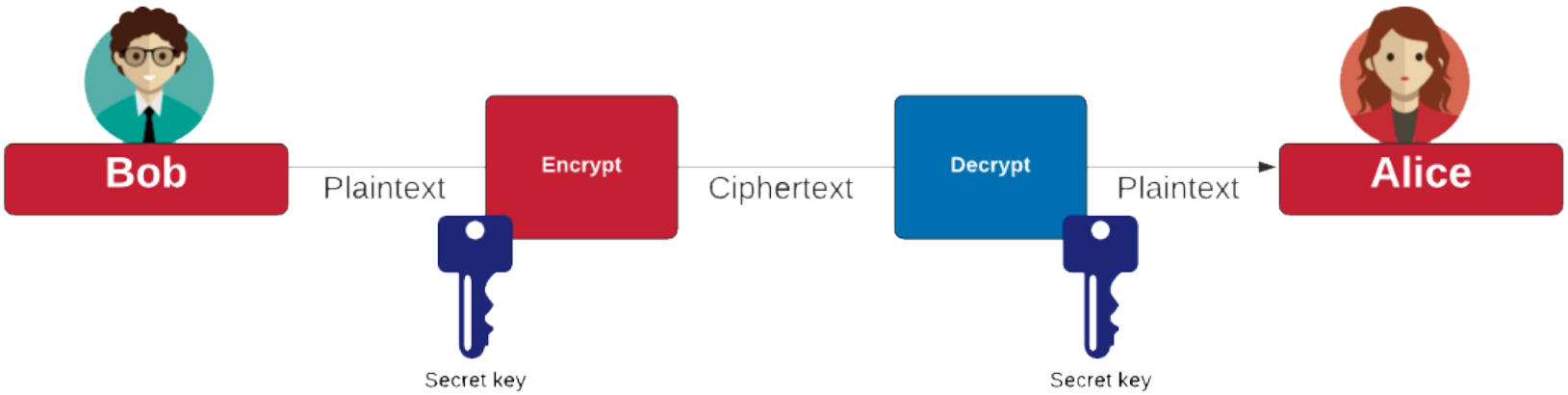
Key
interchange

Phil
Zimmerman



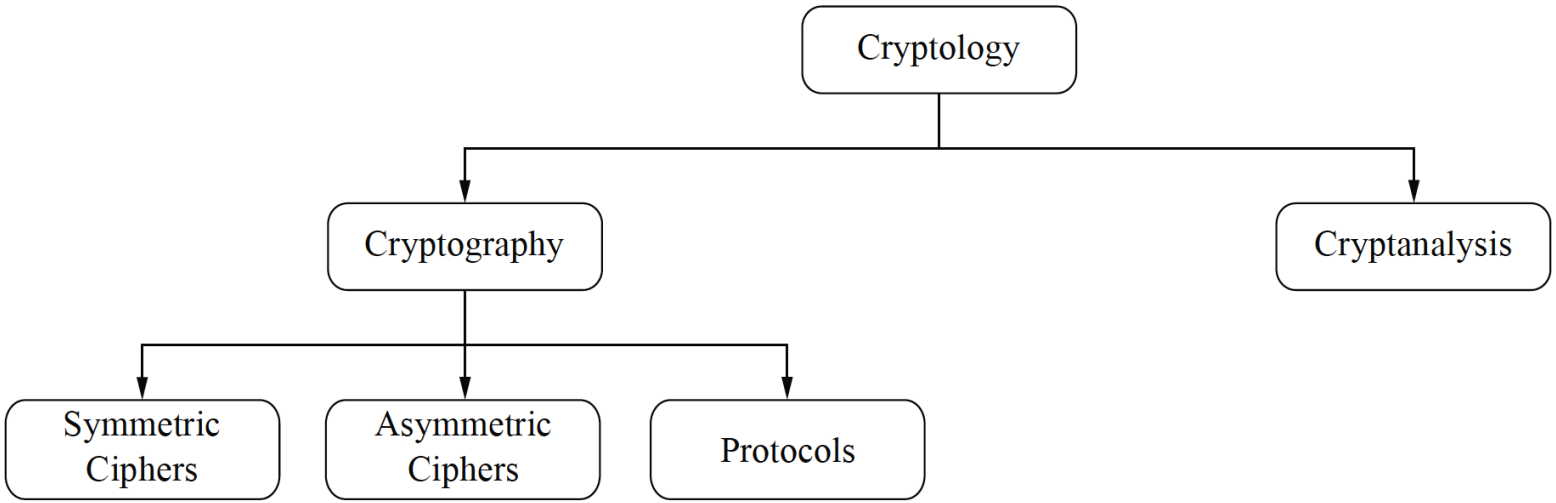
PGP
Encryption

Limitations of Symmetric Cryptography



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Introduction to Public-Key Cryptography

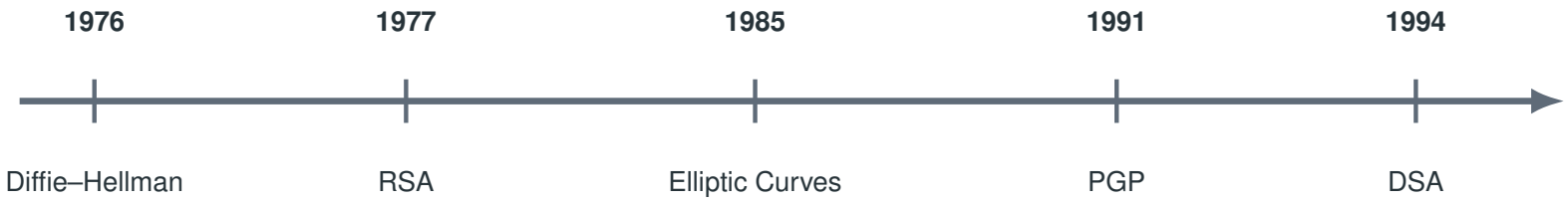


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Evolution of Public-Key Cryptography

1976: New concept
Public-key cryptography and secure key establishment

Digital signatures and public-key systems become practical security tools



Main families:
RSA → related to integer factorization
ECC → elliptic-curve discrete logarithm

Historical Impact

Public-key cryptography enabled **key establishment, encryption, and digital signatures** without requiring a previously shared secret key. Public keys still need an authentication mechanism to prevent impersonation.

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The Discrete Logarithm Problem

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Review of the Discrete Logarithm Problem

Cyclic Group $\mathbb{Z}_{11}^*$

Consider the multiplicative group $\mathbb{Z}_{11}^* = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$. The order of the group is

$$|\mathbb{Z}_{11}^*| = 10.$$

An element $\alpha \in \mathbb{Z}_{11}^*$ is a **primitive element** or **generator** if

$$\boxed{\text{ord}(\alpha) = |\mathbb{Z}_{11}^*|}$$

For $\alpha = 2$,

$$\text{ord}(2) = 10.$$

Therefore, $\alpha = 2$ is a generator of $\mathbb{Z}_{11}^*$.

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Review of the Discrete Logarithm Problem

Discrete Logarithm Problem

For any $\beta \in \mathbb{Z}_{11}^*$, find x such that

$$\boxed{2^x \equiv \beta \pmod{11}}$$

Equivalently,

$$\boxed{x = \log_2(\beta) \pmod{10}}$$

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The Discrete Logarithm Problem

DLP in Prime Fields

Discrete Logarithm over $\mathbb{Z}_p^*$

- Let p be a prime and let $\mathbb{Z}_p^*$ be the multiplicative group of order $p-1$.
- Let α be a generator of $\mathbb{Z}_p^*$ and $\beta \in \mathbb{Z}_p^*$.
- The **Discrete Logarithm Problem (DLP)** is to find x such that

$$\boxed{\alpha^x \equiv \beta \pmod{p}}$$

$$x = \log_\alpha(\beta) \pmod{p-1}$$

Computing $\alpha^x \bmod p$ is efficient, but finding x from α and β is computationally hard for large, well-chosen parameters.

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The Discrete Logarithm Problem

Example : DLP in a Prime Order Subgroup

Consider the multiplicative group $\mathbb{Z}_{47}^*$, where $\alpha = 5$ is a generator.

For $\beta = 41$, find the exponent x such that

$$\boxed{5^x \equiv 41 \pmod{47}}$$

Since the order of $\mathbb{Z}_{47}^*$ is 46, the solution is defined as

$$x \pmod{46}.$$

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The Discrete Logarithm Problem

The Discrete Logarithm Problem (DLP) in Prime Fields

DLP over $\mathbb{Z}_p^*$, where p is a prime

Example

- For cryptographic applications, it is often useful to work in a subgroup of large prime order.
- For cryptographic primes $p > 3$, the order $p - 1$ is even. In practice, many systems use a large prime-order subgroup of $\mathbb{Z}_p^*$.

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The Discrete Logarithm Problem

The Discrete Logarithm Problem (DLP) in Prime Fields

DLP in a Prime-Order Subgroup

Example

- Consider a discrete logarithm in the group $\mathbb{Z}_{47}^*$ with order 46.
- Since $46 = 2 \cdot 23$, the proper subgroup orders are 1, 2, and 23. The full group has order 46.
- $\alpha = 2$ has order 23, so it generates the subgroup with 23 elements.
- For $\beta = 36$, find x modulo 23 such that

$$2^x \equiv 36 \bmod 47$$

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The Discrete Logarithm Problem

The Generalized Discrete Logarithm Problem

DLP in a finite cyclic group

Definition

- Let $G = \langle \alpha \rangle$ be a finite cyclic group of order n, and let $\beta \in G$.
- The DLP is to find x modulo n such that, in multiplicative notation,

$$\beta = \alpha^x.$$

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The Discrete Logarithm Problem

Discrete Logarithm in an Additive Group

- Consider the additive group $(\mathbb{Z}_{11}, +)$.
- This is a cyclic group, and $\alpha = 2$ is a generator.
- α generates the group:

i	1	2	3	4	5	6	7	8	9	10	11
$i \cdot \alpha$	2	4	6	8	10	1	3	5	7	9	0

- For $\beta = 3$, find x modulo 11 such that

$$x \cdot 2 = \underbrace{2 + 2 + \cdots + 2}_{x \text{ times}} \equiv 3 \pmod{11}$$

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The DLP in Different Groups

Multiplicative Notation

$$\boxed{\beta = \alpha^x}$$

Example:

$$\alpha^x \equiv \beta \pmod{p}.$$

Additive Notation

$$\boxed{Q = xP}$$

Here, xP means
$P + P + \cdots + P$, repeated x times.

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The Discrete Logarithm Problem

Groups Used in Discrete-Logarithm Cryptography

- A large prime-order cyclic subgroup of the multiplicative group $\mathbb{F}_p^*$
  - Diffie–Hellman Key Exchange (DHKE)
  - ElGamal Encryption
  - Digital Signature Algorithm (DSA)
- A large prime-order cyclic subgroup of the elliptic-curve group $E(\mathbb{F}_p)$
  - Elliptic-Curve Diffie–Hellman (ECDH)
  - Elliptic-Curve Digital Signature Algorithm (ECDSA)
  - Elliptic-Curve Integrated Encryption Scheme (ECIES)

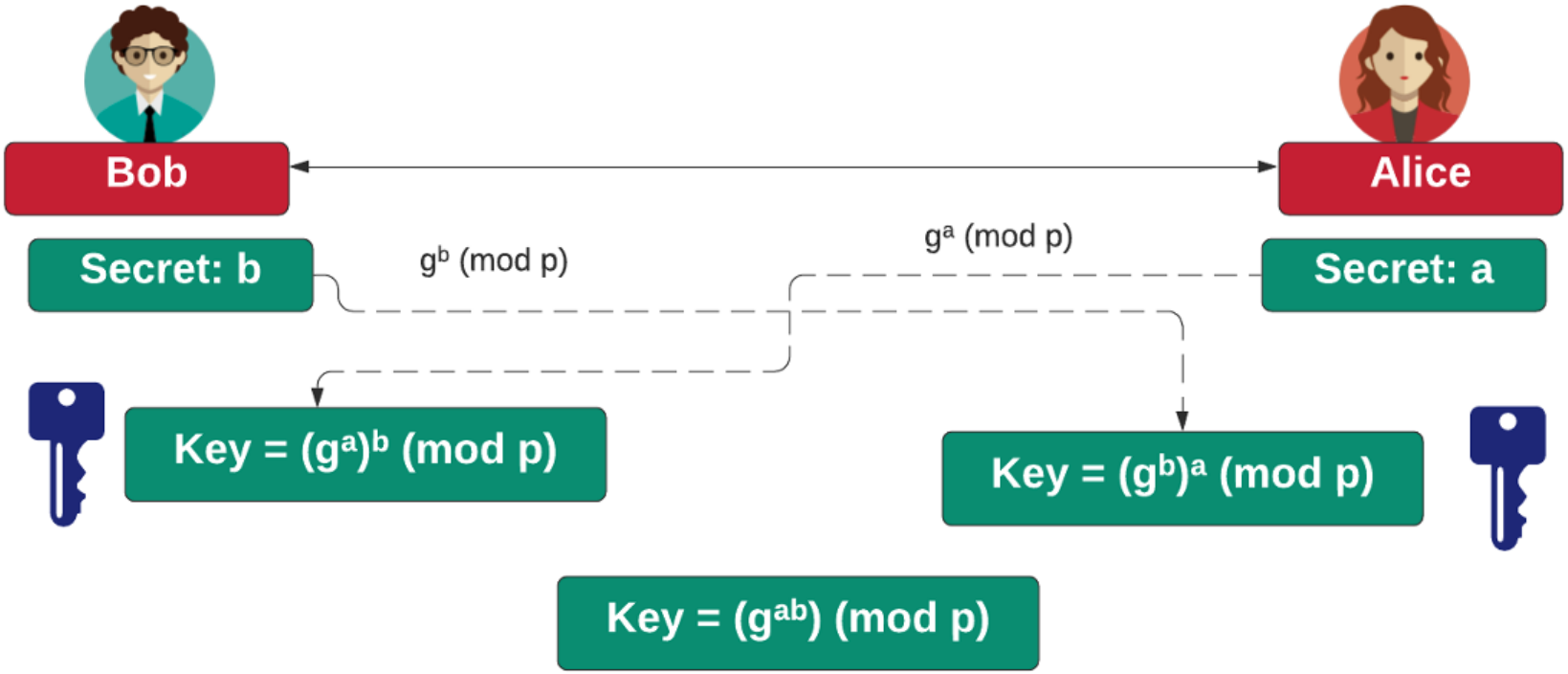
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DLP and Diffie–Hellman

DLP and Diffie–Hellman (cont.)

choose a random secret exponent b | **(Public)** | **choose a random secret exponent** a

compute public value $B \equiv g^b \pmod{p}$ | (p, g) | **compute public value** $A \equiv g^a \pmod{p}$



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Computational Diffie–Hellman Problem

An attacker knows g, g^a, g^b. The objective is to compute:

$$\boxed{g^{ab}}$$

without knowing a or b.

If an attacker can solve the DLP,

$$g^a \longrightarrow a,$$

then the attacker can compute

$$(g^b)^a = g^{ab}.$$

Therefore,

Solving the DLP breaks Diffie–Hellman

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Attacks

Attack 1: Brute Force

Suppose we want to solve

$$\alpha^x = \beta.$$

The simplest attack computes

$$\alpha^0, \alpha^1, \alpha^2, \ldots$$

until

$$\alpha^x = \beta.$$

Cost

For a group of order n, the attacker may need approximately

$$\boxed{O(n)}$$

group operations.

For a large group, exhaustive search is not practical.

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Square-Root Attacks

Brute force is not the best generic attack.
Better algorithms can solve the DLP in approximately

$$\boxed{O(\sqrt{n})}$$

group operations.

Important examples are

- **Baby-Step Giant-Step**
- **Pollard's Rho for Discrete Logarithms**

If

$$n \approx 2^k,$$

then a generic square-root attack needs roughly

$$\boxed{2^{k/2}}$$

operations.

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Attack 2: Baby-Step Giant-Step

An attacker wants to solve $\beta = \alpha^x$ in a cyclic group of order n.

Choose $m = \lceil \sqrt{n} \rceil$ and write the unknown exponent as

$$x = im + j,$$

where

$$0 \leq i, j < m.$$

Then

$$\alpha^{im+j} = \beta$$

can be rewritten as

$$\boxed{\alpha^j = \beta(\alpha^{-m})^i}$$

The attack looks for a collision between these two expressions.

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Baby-Step Giant-Step: How It Works

1. Baby steps
Compute and store $\alpha^0, \alpha^1, \alpha^2, \ldots, \alpha^{m-1}$.

2. Giant steps
Compute $\beta,\ \beta\alpha^{-m},\ \beta\alpha^{-2m}, \ldots$
until a value matches one of the baby steps.

If $\alpha^j = \beta(\alpha^{-m})^i$, then

$$\boxed{x = im + j}.$$

Cost

Time complexity: $\boxed{O(\sqrt{n})}$

Memory complexity: $\boxed{O(\sqrt{n})}$

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Exercise: Baby-Step Giant-Step

Use the Baby-Step Giant-Step algorithm to solve the discrete logarithm

$$2^x \equiv 15 \pmod{29}.$$

Assume that 2 generates a cyclic group of order 28.

Tasks

1. Compute m
2. Build the baby-step table
3. Generate the giant steps
4. Find a collision between a baby step and a giant step.
5. Recover the exponent

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Solution: Baby-Step Giant-Step (cont.)

Solve

$$2^x \equiv 15 \pmod{29}.$$

Since the group order is $n = 28$,

$$m = \left\lceil \sqrt{28} \right\rceil = 6.$$

Baby steps:

j	0	1	2	3	4	5
$2^j \bmod 29$	1	2	4	8	16	3

Since

$$2^6 \equiv 6 \pmod{29},$$

we need

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Solution: Baby-Step Giant-Step (cont.)

$$2^{-6} \equiv 6^{-1} \equiv 5 \pmod{29}.$$

Giant steps:

i	0	1	2	3	4
$15(2^{-6})^i \bmod 29$	15	17	27	19	8

We find the collision

$$8 = 2^3 = 15(2^{-6})^4.$$

Therefore,

$$x = im + j = (4)(6) + 3 = \boxed{27}.$$

Finally,

$$\boxed{2^{27} \equiv 15 \pmod{29}}.$$

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Attack 3: Pollard's Rho for DLP

Suppose we want to solve $\beta = \alpha^x$ in a cyclic group of order n.
Pollard's Rho generates values of the form $X_i = \alpha^{a_i}\beta^{b_i}$.
The goal is to find a **collision**:
$X_i = X_j$.
Since $\beta = \alpha^x$,
$\alpha^{a_i+b_i x} = \alpha^{a_j+b_j x}$.
Therefore,
$(b_i - b_j)x \equiv a_j - a_i \pmod{n}$.
If $(b_i - b_j)$ has an inverse,

$$\boxed{x \equiv (a_j - a_i)(b_i - b_j)^{-1} \pmod{n}}.$$

Cost

Time: $\boxed{O(\sqrt{n})}$ Memory: $\boxed{O(1)}$

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Exercise: Pollard's Rho for DLP

Solve $2^x \equiv 13 \pmod{23}$.
The element $\alpha = 2$ generates a subgroup of order $n = 11$.
Use
$X_i = 2^{a_i} 13^{b_i} \pmod{23}$.
Start with
$(X_0, a_0, b_0) = (2, 1, 0)$.
Use the rule

$X_i \bmod 3$	Next value
0	$X_{i+1} = 13X_i$
1	$X_{i+1} = X_i^2$
2	$X_{i+1} = 2X_i$

Update (a_i, b_i) using the same operation.
Task: Find a collision $X_i = X_j$ and recover x.

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Solution: Pollard's Rho for DLP (cont.)

The generated values are

i	X_i	a_i	b_i
0	2	1	0
1	4	2	0
2	16	4	0
3	3	8	0
4	16	8	1
5	3	5	2

We find the collision $X_3 = X_5 = 3$.
Thus,
$2^8 = 2^5\,13^2$.
Since $13 = 2^x$,
$8 \equiv 5 + 2x \pmod{11}$,

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Solution: Pollard's Rho for DLP (cont.)

so
$2x \equiv 3 \pmod{11}$.
Because $2^{-1} \equiv 6 \pmod{11}$,
$x \equiv 3(6) \equiv \boxed{7} \pmod{11}$.
Verification:
$\boxed{2^7 \equiv 13 \pmod{23}}$.

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Security of Diffie–Hellman

For Diffie–Hellman to be secure, the selected group must make the Discrete Logarithm Problem difficult.

The attacker observes

$$\alpha, \qquad A = \alpha^a, \qquad B = \alpha^b.$$

The secret values are

$$a, \qquad b.$$

Recovering a requires solving

$$\boxed{a = \log_\alpha(A)}$$

and recovering b requires solving

$$\boxed{b = \log_\alpha(B).}$$

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Main Lessons

- The DLP is defined in a **cyclic group**.
- The classical DLP in $\mathbb{Z}_p^*$ is only one particular case.
- The generalized DLP allows us to study the problem in other groups.
- Generic attacks require about $\sqrt{n}$ group operations.
- Some groups allow additional attacks such as index calculus.

The choice of the group determines the security.

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Lab 5. Programming Attacks on the Discrete Logarithm Problem (cont.)

Objective

Implement and compare three algorithms to solve the Discrete Logarithm Problem

$$\alpha^x \equiv \beta \pmod{p}.$$

1. **Brute Force**
   Test successive values of x until

   $$\alpha^x \equiv \beta \pmod{p}.$$

2. **Baby-Step Giant-Step**
   Use the time–memory trade-off with approximately

   $$O(\sqrt{n})$$

   operations.

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Lab 5. Programming Attacks on the Discrete Logarithm Problem (cont.)

3. **Pollard's Rho for DLP**
   Use a pseudo-random sequence and collision detection to recover x.

Tasks

For the same values of (p, α, β):

- Implement the three algorithms.
- Verify that all methods recover the same x.
- Measure execution time.
- Compare memory usage and computational cost.
- Analyze how the execution time changes as p increases.

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